

Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

I need some help!



Learning Outcome:

Talk about principles, concepts, and characteristics of general-purpose computer programming language in spoken and written situations.

Can Do Statements:

- I can say what the Boolean data type is.
- I can talk about the syntax to use in the Boolean data type.

Ask your partner:

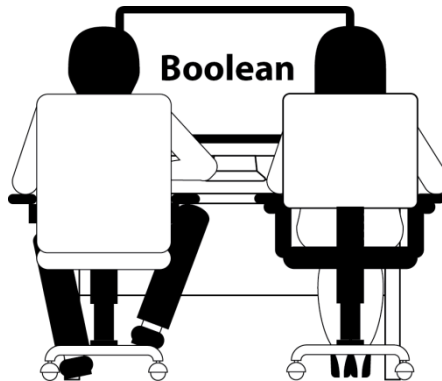
- Do you recognize the term "Boolean"? In which context have you heard about it?
- Why do you think the term primitive is used to describe a data type?





Let's Explore!

John and Catherine are talking about Boolean Data.



Instructions: Read the text below.

- John:** Hi Cat! I'm studying for the exam, but I'm a little bit lost. Have I got to study the Boolean data type?
- Catherine:** You'd better do it! I see you're really lost! You've got to have it clear if you really want to pass that exam. You might want to start by understanding what it is.
- John:** Sure! That would be a good start!
- Catherine:** The Boolean data type is one of the primitive data types supported by Java that allows you to represent logical values. It is said to be primitive because it comes within the programming language.
- John:** I see. Let me confirm something. This Boolean logic is also embedded in the *if-statements*, isn't it?
- Catherine:** Yes, it is! Let's continue, this data type only supports two possible values that are *true* and *false* represented by *1* and *0*, respectively. You've got to have clear that these both are just keywords the programming language use; they are not destined for any variable, function or object name. If you're going to do some research, you'll probably find something about the concept of a *switch*, which is a core concept to refer to these *true* and *false* terms.
- John:** Well, I think I get it. Do you think I am going to have to study the syntax of this data type?
- Catherine:** Absolutely! You ought to know how to use it if you want to fully understand the concept! There are three structures you should have in mind. The first variable is ***public boolean isValid()***. You can use this one for a remote operation; however, it is not recommended to use it later in your code since Java wouldn't know what value you are trying to establish. The other two variables are ***boolean flag= true;*** and ***Boolean stop= false;***. These are used depending on what you want to do with the code.
- John:** Am I going to have to memorize all of this?
- Catherine:** Well, you might want to search for examples and practice a lot! That's a good way to learn this by heart.



Let's Practice!

Define this words before taking the exam.

Instructions: Match column A with column B. Read the sentences aloud. Check your pronunciation with your instructor.

COLUMN A

1. a primitive data type in Java
2. a function representing a number or value
3. a word that decrypts a code
4. the way in which something is written in computer language a basic data type that is given by a programming language

COLUMN B

- a. Syntax
- b. Keyword
- c. Primitive data type
- d. Boolean
- e. Variable



Let's Practice!

Listen to some pieces of advice! Is syntax really important?

Instructions: Select the best options in the table. There is only one answer per sentence.





Let's Practice!

	<i>public boolean isValid().</i>	statement. This is one of the most fundamental things you'll use in Java.	the Boolean data type has only two values.	Remember that all variable's data types and keywords have to be declared first.	try it yourself. Type boolean in lower case, type a keyword and equals.
You might want to create an if-					
You ought to remember that					
You might want to practice how to use the first variable that is					
You'd better follow the recommendations on how to declare keyword and variables.					
You might want to					



Extension Activity

Describing the Syntax in Boolean Data Type

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for information about examples of Boolean data types.
- Write down information and describe how the syntax works.
- Create a poster to explain each one of the elements in the syntax.
- Share with your instructor.

Self-Assessment

Yes

No

- I look for information about examples of Boolean data types.
- I describe how the syntax works.
- I create a poster to explain each one of the elements in the syntax.
- I talk about Boolean data syntax.

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Target Words

- **keyword** (noun): a word that decrypts a code
e.g. You've got to have clear that these both are just keywords the programming language use.
- **variable** (noun): a function representing a number or value
e.g. The first variable is **public boolean isValid()**.
- **primitive data type** (noun): a basic data type that is given by a programming language
e.g. The Boolean data type is one of the primitive data types supported by Java that allows you to represent logical values.
- **learn by heart** (expression): to memorize
e.g. Well, you might want to search for examples and practice a lot! That's a good way to learn this by heart.
- **syntax** (noun): the way in which something is written in a computer language
e.g. The syntax for the IF-statement is: if a condition is true, follow the code inside the curly brackets
- **embed** (verb): to include something as an essential part of something else
e.g. This Boolean logic is also embedded in the if-statements, isn't it?



Learning Points

Have got to and be going to have to for what is necessary in questions.

We use **have got to** or **have to** when we want to express strong obligation or the need to do something.

Have got to and **have to** are very similar in meaning. The difference between them consists in the level of formality you want to give to your speech. **Have got to** is used in informal speech, while **have to** is more formal. When **have to** is accompanied by **be going to**, it means that the fulfillment of the action represents an obligation in the future. On the other hand, **have got to** is only used in simple present tense.

Let's look at the following structure to ask questions with these grammar constructions:

Be+ subject+ going to have to + verb+ complement+?

*Are you **going to have to** check if-statements in Java?*

Wh questions+ be + subject+ going to have to + verb+ complement+?

*What are you **going to have to check** to understand if statements?*

In American English, it is not very common to use **have got to** in questions; you can use **have to** instead. However, look at the following structure:

Have/has+ subject+ got to + verb+ complement+?

*Have you **got to learn** what an if-statement is?*

Do/Does + subject+ have/has to + verb+ complement?

*Do you **have to learn** what an if-statement is?*

Examples

- *Do I **have to** study the Boolean data type?*
- *Do you think I am **going to have to** study the syntax of this data type?*
- *You've **got to have** clear that these both are just keywords the programming language use.*
- *Am I **going to have to** memorize all of this?*



Learning Points

Had Better, Ought to, and Might Want to for What Is Advisable

Had better, ought to and might want to are modal verbs you use to give advice.

When you are sure there'll be a negative result if someone doesn't do what is expected, use **had better** (also **'d better**) to express a strong piece of advice.

For example:

You'd better use the lower case in Java. (If you don't, something bad will happen.)

Ought to is similar to **should**; it is used to express what is necessary as a piece of advice. The difference between them is that **ought to** can't be used in questions, and it is more formal. **Should** is more common, though.

For example:

By now, you ought to know all the data types by heart. (By now, you should know all the data types by heart.)

Might want to, for its part, is a polite way of suggesting something as a good option.

For example:

If you are going to be in charge of that task, you might want to fully understand the Boolean data type.

Examples

- *You'd better study the Boolean data type!*
- *You might want to start by understanding what the Boolean data type is.*
- *You ought to know how to use it if you want to fully understand the concept!*